Constructing Machine-Precision Neural Networks with Quasi-Interpolants

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Abstract

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Our goal is to build an interpolation scheme with arbitrarily good approximation that is also expressible in the primitives available to one-hidden-layer MLPs. Quasi-interpolants are a natural choice because they reconstruct from samples using translates of a single cardinal function, which will later construct directly with hidden units and activations. We first construct the quasi-interpolant and its cardinal function here, then realize it as an MLP (Section 3.2), derive the error-width tradeoff (Section 3.3), and summarize the implementation in Section 3.4.

Main result (informal). For any analytic target f on $[-1, 1]$ and any target precision $\varepsilon > 0$ above a kernel-dependent threshold, we construct a one-hidden-layer MLP with $W = O(\log(1/\varepsilon))$ neurons and maximum weight magnitude $O(\log(1/\varepsilon))$ that interpolates f to accuracy ε in L^∞. The construction is explicit (Algorithm 1); for the sech^2 kernel, the viable parameter regime extends to fp64 machine precision ($\varepsilon \sim 10^{-15}$), with all weights remaining within floating-point range (Table 1).

3.1 Constructing Fourier-Normalized Quasi-Interpolants

Setup. To begin our Quasi-interpolation construction from MLPs, we let $\Omega = [-1, 1]$, fix $N \in \mathbb{N}$, and define

$$h := \frac{2}{N}, \qquad x_k := -1 + kh, \qquad K_\gamma(x) := \gamma K(\gamma x). \tag{1}$$

A translation-invariant quasi-interpolant reconstructs f from uniform samples as

$$(Q_h f)(x) := \sum_{k \in \mathbb{Z}} f(x_k)\, L_h(x - x_k), \tag{2}$$

where the ideal *cardinal function* L_h would satisfy the following two conditions

- **(i)** $\boldsymbol{L_h(jh) = \delta_{j,0}}$ for all $j \in \mathbb{Z}$, ensuring exact interpolation at grid points.
- **(ii)** $\boldsymbol{L_h(x) = \sum_j c_j\, K_\gamma(x - jh)}$, so the operator maps onto a one-hidden-layer MLP (Section 3.2).

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Furthermore, stability between nodes is guaranteed by the kernel's exponential spatial decay (Assumption K4, Appendix A.1), which makes L_h a localized bump, the quasi-interpolation analog of avoiding the Runge phenomenon.

Fourier normalization. By Appendix B.1, condition **(i)** is equivalent to the Fourier partition-of-unity condition

$$\sum_{m\in\mathbb{Z}} \widehat{L}_h\left(\omega + \frac{2\pi m}{h}\right) = h \qquad \forall\, \omega. \tag{3}$$

And enforcing the structure found in condition **(ii)** lets $\widehat{L}_h(\omega)$ be decoupled via shift invariance, giving us

$$\widehat{L}_h(\omega) = \widehat{K}_\gamma(\omega) \sum_j c_j e^{-ijh\omega} = \widehat{K}_\gamma(\omega)\, \widehat{C}_h(\omega), \qquad \widehat{C}_h(\omega) := \sum_j c_j e^{-ijh\omega}.$$

Since $\widehat{C}_h$ is $2\pi/h$-periodic, plugging this into (3) gives

$$h = \widehat{C}_h(\omega) \sum_{m\in\mathbb{Z}} \widehat{K}_\gamma\left(\omega + \frac{2\pi m}{h}\right) = \widehat{C}_h(\omega)\, D_h(\omega),$$

so the construction reduces to a single division in Fourier space:

$$\widehat{C}_h(\omega) = \frac{h}{D_h(\omega)}, \qquad D_h(\omega) := \sum_{m\in\mathbb{Z}} \widehat{K}_\gamma\left(\omega + \frac{2\pi m}{h}\right). \tag{4}$$

Here $\widehat{C}_h$ is the *Fourier character* and D_h is the *normalizer*, the periodized kernel spectrum. The construction requires $D_h \neq 0$ everywhere, which we verify for $K = \frac{1}{2}\operatorname{sech}^2$ via diagonal dominance (Appendix A.1, Proposition 4). Rescaling $\theta = h\omega$ gives $\widetilde{C}_\lambda(\theta) := \widehat{C}_h(\theta/h) = \sum_j c_j e^{-ij\theta}$, so the Fourier coefficients of $\widetilde{C}_\lambda$ are exactly the same coefficients c_j appearing in the original kernel expansion of L_h. In practice, these coefficients are computed using the DFT shown explicitly in Algorithm 1.

From infinite to finite. To implement the operator on Ω, we introduce two truncations. *Stencil truncation*: restrict the cardinal function to $|j| \le K_c$ terms,

$$L_h^{(K_c)}(x) := \sum_{|j|\le K_c} c_j\, K_\gamma(x - jh). \tag{5}$$

Halo truncation: restrict the lattice sum to $k \in \{-R, \dots, N+R\}$, placing R nodes on each side of Ω to stabilize boundary effects (Figure 2a). The finite operator is

$$(Q_{h,R}^{(K_c)} f)(x) := \sum_{k=-R}^{N+R} f(x_k)\, L_h^{(K_c)}(x - x_k), \qquad x \in \Omega, \tag{6}$$

which re-indexes into a *kernel network* (Appendix B.3):

$$(Q_{h,R}^{(K_c)} f)(x) = \sum_{m=-R}^{N+R} a[m]\, K_\gamma(x - x_m), \qquad a[m] := \sum_{|j|\le K_c} c_j\, f(x_{m-j}). \tag{7}$$

3.2 MLPs Can Implement Quasi-Interpolants

The kernel network (7) uses K_γ as its basis, but standard MLP activations ψ (e.g., $\tanh$) typically do not satisfy the kernel requirements directly. Instead, a low-order derivative often does: $\operatorname{sech}^2 = \tanh'$. Assume

$$K(u) = \psi^{(r)}(u) \tag{8}$$

for some smooth ψ and order $r \in \mathbb{N}$, with K satisfying the kernel conditions (Assumptions K1–K4). By the chain rule, $K_\gamma(x - x_m) = \gamma^{-(r-1)} \frac{d^r}{dx^r} \psi(\gamma(x - x_m))$, so (7) becomes

$$(Q_{h,R}^{(K_c)} f)(x) = \frac{d^r}{dx^r}\left(\sum_{m=-R}^{N+R} \frac{a[m]}{\gamma^{r-1}}\, \psi\big(\gamma(x - x_m)\big)\right). \tag{9}$$

Define the one-hidden-layer MLP

$$g_{\mathrm{MLP}}(x) := \sum_{m=-R}^{N+R} w[m]\,\psi\big(\gamma(x - x_m)\big), \qquad w[m] := \frac{a[m]}{\gamma^{r-1}}. \tag{10}$$

Then

$$g_{\mathrm{MLP}}^{(r)}(x) = (Q_{h,R}^{(K_c)} f)(x). \tag{11}$$

Applying the construction to $f^{(r)}$, the MLP recovers f up to an integration polynomial:

$$\widetilde{f}(x) = p_{r-1}(x) + g_{\mathrm{MLP}}(x), \tag{12}$$

where p_{r-1} is fixed by boundary conditions. For $\tanh$ ($r = 1$, $K = \mathrm{sech}^2$), all kernel conditions are verified (Appendix A.1) and the polynomial reduces to a bias. Extending the construction to other activations (e.g., GELU, Swish) via higher-order derivatives is possible in principle but requires separate verification of the kernel conditions for each case.

Expressivity is not the bottleneck. Equations (10)–(12) show that one-hidden-layer tanh MLPs can exactly represent the quasi-interpolant construction. Whether machine-precision interpolation is achievable reduces to whether the correct weights can be found, an optimization question.

3.3 Error Analysis and Width-Precision Scaling

The dimensionless bandwidth. We now introduce the dimensionless bandwidth parameter

$$\lambda := \gamma h.$$

Since the kernel width is on the order of $1/\gamma$ and the grid spacing is h, the quantity $1/\lambda = 1/(\gamma h)$ can be interpreted as the approximate number of grid points that lie inside one kernel width. The parameter λ controls a fundamental tradeoff. When λ is large (narrow kernels), the cardinal function is easy to construct but the kernel's spectral energy above the Nyquist rate π/h folds back into D_h, producing *aliasing error*. When λ is small (wide kernels), aliasing vanishes but building a cardinal function from broad overlapping bumps demands large coefficients with precise cancellations, making the construction *ill-conditioned*, reflected in error-bound prefactors that diverge algebraically as $\lambda \to 0$ (Theorem 1). The optimal λ balances these effects.

Error decomposition. The total error decomposes into four terms by the triangle inequality (full proof in Appendix A.1, Theorem 5).

Theorem 1 (Numerical error bound). *Let $\lambda = \gamma h$. Assume f is analytic in a strip of half-width ρ around Ω, and assume the normalizer D_h is bounded away from zero on a strip of half-width σ. Then*

$$\|f - Q_{h,R}^{(K_c)} f\|_{L^\infty(\Omega)} \le \underbrace{C_1(\lambda)\, e^{-c_1/h}}_{\text{resolution}} + \underbrace{C_2\, e^{-c_2/\lambda^p}}_{\text{aliasing}} + \underbrace{C_3(\lambda)\, e^{-c_3\lambda R}}_{\text{halo}} + \underbrace{C_4(\lambda)\, e^{-c_4\lambda K_c}}_{\text{stencil}}. \tag{13}$$

The rates $c_1, \ldots, c_4$ depend on the target's analyticity ρ, the kernel's decay rate a_K, and the normalizer's strip margin σ. The prefactors C_1, C_3, C_4 capture the stability cost of Fourier normalization and diverge algebraically as $\lambda \to 0$ (explicit formulas in Appendix A.1, Eqs. 29–31). C_2 is independent of λ. For $K = \mathrm{sech}^2$, $p = 1$.

Each term is exponential in a parameter we control. The resolution term decays as $e^{-c_1 N/2}$ with the number of grid points; the aliasing term decays as e^{-c_2/λ^p}, vanishing as the kernel widens relative to the grid (small λ); and the halo/stencil terms decay geometrically in R and K_c, respectively, at rates coupled through λ.

Width-precision scaling. To achieve a target tolerance ε, we choose λ in the *viable regime*: small enough that $C_2 e^{-c_2/\lambda^p} \le \varepsilon$ (bounding aliasing), yet large enough that the prefactors $C_i(\lambda)$ remain moderate (bounding ill-conditioning). For a fixed precision target such as fp64, this viable regime is a computable interval $[\lambda_{\min}, \lambda_{\max}]$ depending on ε, the kernel, and the analyticity width

Table 1: Comparison of parameter count, working-precision bit complexity, and weight magnitudes for prior MLP constructions on analytic targets.

Method	#params vs. ε	Bit complexity (working precision)	Weight magnitudes
Cybenko (1989)	$\Theta(\varepsilon^{-1})$	$\Theta(\log(1/\varepsilon))$	$\|W\|_{\max} = \Theta\big(\varepsilon^{-1}\log(1/\varepsilon)\big)$
Mhaskar (1993)	$\widetilde{O}\big(\log(1/\varepsilon)\log\log(1/\varepsilon)\big)$	$\gtrsim \widetilde{\Theta}\big(\log(1/\varepsilon)\log\log(1/\varepsilon)\big)$	$\|W\|_{\max} = \Theta(\varepsilon^{-1/k})$ (order k)
Quasi-interpolant (ours)	$O(\log(1/\varepsilon))$	$O(\log(1/\varepsilon) + \log\log(1/\varepsilon))$	$\|W\|_{\max} = O(\log(1/\varepsilon))$

Algorithm 1 Explicit Quasi-Interpolant MLP Construction

Input: grid size N, bandwidth γ, halo R, stencil half-width K_c, DFT size M, activation ψ with $K = \psi^{(r)}$, and target samples on the extended grid.

1: Build the extended uniform grid on $[-1, 1]$ together with the halo points needed for the finite operator.
2: Sample the target f on this extended grid.
3: Evaluate the periodized kernel spectrum D_h on a DFT grid and form the Fourier character $\widetilde{C}_\lambda = h/\widetilde{D}_\lambda$.
4: Recover the cardinal coefficients c_j by inverse DFT, and retain the stencil coefficients with $|j| \le K_c$.
5: Form the truncated cardinal function $L_h^{(K_c)}(x) = \sum_{|j|\le K_c} c_j K_\gamma(x - jh)$.
6: Convolve the stencil coefficients c_j with the sampled target values to obtain the kernel-network coefficients $a[m]$.
7: Convert the kernel network into MLP parameters by using hidden units $\psi(\gamma(x - x_m))$ and outer weights $w[m] = a[m]/\gamma^{r-1}$.
8: If $r > 1$, determine the degree-$(r-1)$ integration polynomial from boundary conditions.
9: Output the final approximant $\widetilde{f}(x) = p_{r-1}(x) + \sum_{m=-R}^{N+R} w[m]\psi(\gamma(x - x_m))$.

Output: the explicit one-hidden-layer MLP $\widetilde{f}$.
Width: $W = N + 2R + 1$ neurons, plus r polynomial coefficients when $r > 1$.

(Appendix C.1). With any λ^* in this interval, the remaining three terms each require their control parameter to scale as $O(\log(1/\varepsilon))$:

$$N = O\left(\frac{1}{c_1}\log\frac{1}{\varepsilon}\right), \qquad R = O\left(\frac{1}{c_3\lambda^*}\log\frac{1}{\varepsilon}\right), \qquad K_c = O\left(\frac{1}{c_4\lambda^*}\log\frac{1}{\varepsilon}\right).$$

Since the MLP has $W = N + 2R + 1$ neurons, the total width scales as $W = O(\log(1/\varepsilon))$. Equivalently,

Corollary 1 (Exponential convergence in width). *Fix a target precision ε in the viable regime (Appendix C.1) and choose λ^* accordingly. There exist $A, \alpha > 0$ such that the constructed MLP with W neurons satisfies $\|f - \widetilde{f}_W\|_{L^\infty} \le A\, e^{-\alpha W}$. (Proof in Appendix C.2.)*

This matches the rate of classical polynomial interpolation for analytic targets (Trefethen, 2019), realized by a one-hidden-layer MLP with explicit weights.

Weight scaling. Maintaining λ^* constant as $h \to 0$ requires $\gamma = \lambda^*/h = \lambda^* N/2$: the inner-layer weights must grow proportionally with width. The maximum weight magnitude is $O(W) = O(\log(1/\varepsilon))$, keeping all parameters within fp64 range even at machine precision (Appendix C.3). This is what makes the construction numerically realizable, in contrast to prior results where weights scale as $\Theta(\varepsilon^{-1})$ (Cybenko, 1989) or $\Theta(\varepsilon^{-1/k})$ (Mhaskar, 1993); see Table 1.

Predictions for trained networks. The construction makes three testable predictions about what a trained MLP *should* look like to achieve high precision: (i) $\lambda = \gamma h$ should plateau as width increases, requiring γ to grow proportionally with N; (ii) the outer-layer weight magnitudes should remain $O(1)$, not grow; and (iii) the learned features should distribute approximation power uniformly across neurons rather than concentrating it in a compressible subset. In Section 4, we verify that end-to-end trained MLPs violate all three: γ stays $O(1)$ (driving $\lambda \to 0$), outer weights are too large, and learned representations exhibit rank saturation.

3.4 Algorithm

4 Experiments

5 Discussion

References

A Theoretical Results

B Construction Derivations

This appendix derives the Fourier normalization construction summarized in Section 3.1. The proof of Theorem 1 (with explicit constants), Lemmas 2–4, and the kernel verification for sech^2 are in Appendix A.1.

B.1 Deriving the Cardinal Function via Fourier Normalization

We establish that the cardinal property $L_h(jh) = \delta_{j,0}$ is equivalent to (3), and show how Fourier normalization yields equation (4).

B.1.1 The cardinal property in Fourier space

Write

$$L_h(jh) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{L}_h(\omega)\, e^{i\omega jh}\, d\omega.$$

Partition $\mathbb{R}$ into intervals of length $2\pi/h$ and substitute $\omega \mapsto \omega + 2\pi m/h$ in each:

$$L_h(jh) = \frac{1}{2\pi} \int_{-\pi/h}^{\pi/h} \underbrace{\left[\sum_{m\in\mathbb{Z}} \widehat{L}_h\left(\omega + \frac{2\pi m}{h}\right)\right]}_{=:\, S(\omega)} e^{i\omega jh}\, d\omega,$$

using $e^{i(2\pi m/h)jh} = 1$. The integral extracts the j-th Fourier coefficient of S on $[-\pi/h, \pi/h]$.

Forward. If $S(\omega) = h$, then

$$L_h(jh) = \frac{h}{2\pi} \int_{-\pi/h}^{\pi/h} e^{i\omega jh}\, d\omega = \delta_{j,0}$$

by orthogonality.

Reverse. If $L_h(jh) = \delta_{j,0}$ for all j, all nonzero Fourier coefficients of S vanish, forcing $S = h$.

B.1.2 From kernel composition to equation (4)

Write $L_h(x) = \sum_j c_j K_\gamma(x - jh)$. Fourier-transforming gives

$$\widehat{L}_h(\omega) = \widehat{K}_\gamma(\omega)\, \widehat{C}_h(\omega), \qquad \widehat{C}_h(\omega) := \sum_j c_j e^{-ijh\omega}.$$

Because $\widehat{C}_h$ is $2\pi/h$-periodic,

$$\sum_{m\in\mathbb{Z}} \widehat{L}_h\left(\omega + \frac{2\pi m}{h}\right) = \widehat{C}_h(\omega) \sum_{m\in\mathbb{Z}} \widehat{K}_\gamma\left(\omega + \frac{2\pi m}{h}\right) = \widehat{C}_h(\omega)\, D_h(\omega).$$

Plugging this into (3) yields

$$h = \widehat{C}_h(\omega)\, D_h(\omega),$$

so

$$\widehat{C}_h(\omega) = \frac{h}{D_h(\omega)}.$$

B.1.3 Recovering the coefficients c_j

Rescale $\theta := h\omega$, and define

$$\widetilde{C}_\lambda(\theta) := \widehat{C}_h(\theta/h) = \frac{h}{\widetilde{D}_\lambda(\theta)}.$$

This is 2π-periodic with Fourier series

$$\widetilde{C}_\lambda(\theta) = \sum_j c_j e^{-ij\theta}.$$

The Fourier coefficients of $\widetilde{C}_\lambda$ are the kernel-expansion weights c_j.

Remark. The rescaled normalizer $\widetilde{D}_\lambda(\theta) = D_h(\theta/h)$ depends on γ and h only through $\lambda = \gamma h$, which is why λ is the natural dimensionless parameter.

B.2 DFT Approximation of the Cardinal Coefficients

The coefficients c_j are computed via the M-point DFT of $\widetilde{C}_\lambda$ at nodes $\theta_l = 2\pi l/M$ (Algorithm 1, Step 2). The DFT returns aliased coefficients

$$\widetilde{c}_j^{(M)} = c_j + \sum_{l\neq 0} c_{j+lM}.$$

By Lemma 2, $|c_j| \leq C_c(\lambda)e^{-\sigma|j|}$, so

$$\left|\widetilde{c}_j^{(M)} - c_j\right| \leq \frac{2C_c(\lambda)\, e^{-\sigma M}}{1 - e^{-\sigma M}}.$$

This computational error is independent of the four approximation-theoretic terms in Theorem 1 and can be driven below machine precision by choosing M moderately larger than $2K_c + 1$.

B.3 Kernel Network Re-indexing

Substituting (5) into (6) and setting $m = k + j$ gives

$$(Q_{h,R}^{(K_c)} f)(x) = \sum_{k=-R}^{N+R} f(x_k) \sum_{|j|\leq K_c} c_j K_\gamma(x - x_k - jh) = \sum_{m=-R}^{N+R} a[m]\, K_\gamma(x - x_m),$$

where

$$a[m] = \sum_{|j|\leq K_c} c_j f(x_{m-j})$$

is a discrete convolution of width $2K_c+1$. This requires samples on the extended index set $\{-R-K_c, \ldots, N+R+K_c\}$.

B.4 Activation Coverage

The derivative-kernel relation $K = \psi^{(r)}$ is verified for $\tanh$:

Activation ψ	Kernel K	Order r	Integration polynomial
$\tanh$	sech^2	1	bias

For $\tanh$, the kernel $K = \mathrm{sech}^2$ satisfies all conditions (K1–K4) as verified in Appendix A.1.3: nonzero mass, exponential spatial decay with $(A_K, a_K) = (2, 2)$, positive Fourier transform, and a diagonal-dominance condition ensuring a zero-free strip.

Remark on other activations. In principle, any smooth activation whose r-th derivative satisfies K1–K4 yields a valid construction. However, extending the verification beyond $\tanh$ requires care. For instance, $\mathrm{GELU}''(x) = \phi(x)(2 - x^2)$, which changes sign and has nonzero integral only after checking cancellations. $\mathrm{Softplus}'(x) = \sigma(x)$, which does not decay as $x \to +\infty$ and therefore violates K4 as stated. We leave a systematic treatment of activation coverage to future work; all experiments in this paper use $\tanh$ (equivalently, sech^2) activations.

C Width-Precision Scaling

C.1 The Viable Regime for λ

For a fixed target tolerance ε, the bandwidth λ must satisfy competing constraints.

Aliasing (upper bound). $C_2 e^{-c_2/\lambda^p} \le \varepsilon$ requires

$$\lambda \le \lambda_{\max}(\varepsilon) := \left(\frac{c_2}{\log(C_2/\varepsilon)}\right)^{1/p}.$$

Conditioning (lower bound). The prefactors $C_i(\lambda)$ diverge as $\lambda \to 0$. For $K = \frac{1}{2}\operatorname{sech}^2$, the cardinal function's spatial decay (Lemma 4) requires $a_K \lambda > \sigma$, giving $\lambda > \sigma/2$ with $a_K = 2$. The diagonal-dominance condition (Proposition 4) further requires $q(\sigma, \lambda) < 1$, satisfied for $\lambda \gtrsim 1.2$ with $\sigma = 1$.

The viable regime $[\lambda_{\min}, \lambda_{\max}(\varepsilon)]$ is nonempty for all ε above a kernel-dependent threshold; this is the threshold referenced in the informal main result. The threshold arises because $\lambda_{\max}(\varepsilon) \to 0$ as $\varepsilon \to 0$ while $\lambda_{\min}$ is fixed by conditioning, so there exists a smallest achievable ε for each kernel. Note that $\lambda_{\max}(\varepsilon)$ depends on ε only through a $1/\log(1/\varepsilon)$ factor, so the viable regime shrinks slowly as $\varepsilon \to 0$. For fp64 targets ($\varepsilon \sim 10^{-15}$) with the sech^2 kernel, the interval is well-separated and $\lambda^* \approx 1.5$ is near-optimal across target functions in our experiments.

Practical selection. We choose λ^* by a one-dimensional sweep over $[1, 4]$, evaluating the total bound (13) with all prefactors in closed form (Appendix A.1, Eqs. 29–31).

C.2 Proof of Corollary 1

Fix ε and choose λ^* in the viable regime so that the aliasing term satisfies

$$C_2 e^{-c_2/(\lambda^*)^p} =: \delta \le \varepsilon.$$

Given width W, allocate

$$N = \lfloor \beta W \rfloor, \qquad R = \left\lfloor \frac{(1-\beta)W}{2} \right\rfloor, \qquad K_c = R$$

for fixed $\beta \in (0, 1)$. Then $h = 2/N \le 4/(\beta W)$, and:

- *Resolution:*
$$C_1(\lambda^*) e^{-c_1 N/2} \le C_1(\lambda^*) e^{-c_1 \beta W/4}.$$
- *Halo:*
$$C_3(\lambda^*) e^{-c_3 \lambda^* R} \le C_3(\lambda^*) e^{-c_3 \lambda^* (1-\beta) W/4}.$$
- *Stencil:*
$$C_4(\lambda^*) e^{-c_4 \lambda^* R} \le C_4(\lambda^*) e^{-c_4 \lambda^* (1-\beta) W/4}.$$

Setting

$$\alpha := \min\left(\frac{c_1 \beta}{4}, \frac{c_3 \lambda^*(1-\beta)}{4}, \frac{c_4 \lambda^*(1-\beta)}{4}\right)$$

and

$$A := C_1 + C_3 + C_4 + \delta$$

yields

$$\|f - \widetilde{f}_W\| \le A e^{-\alpha W}$$

for W large enough that $N \ge 2$ and $R \ge 1$. □

Remark. The constant A depends on λ^* and hence (weakly) on ε. For any fixed precision target, A is a computable constant. The proportion β can be optimized:

$$\beta^* = \frac{c_3 \lambda^*}{c_1 + c_3 \lambda^*};$$

for typical parameters, $\beta^* \approx 0.6$.

C.3 Weight Magnitude Scaling

With λ^* fixed,

$$\gamma = \lambda^* N/2.$$

Inner-layer weights. $\gamma = O(W) = O(\log(1/\varepsilon))$ by Corollary 1.

Outer-layer weights.

$$|a[m]| \leq \|f\|_\infty \sum_{|j| \leq K_c} |c_j| \leq \|f\|_\infty C_c(\lambda^*) \coth(\sigma/2) = O(1)$$

by Lemma 2. So $w[m] = a[m]/\gamma^{r-1}$: for $r = 1$ (tanh), $w[m] = O(1)$.

Maximum weight.

$$\|W\|_{\max} = \gamma = O(\log(1/\varepsilon)).$$

Concretely, for $\varepsilon = 10^{-15}$: our construction gives $\gamma \sim 35$, versus $\sim 10^{15}$ for Cybenko (1989) and $\sim 10^{15/k}$ for Mhaskar (1993).